2026 Senior Honor Student - Research Project

Studying the chemical evolution of N and O in local Star-forming galaxies using CLASSY and LzLCS

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Duration: January 2026 - March 2026 (~ 10 weeks)

Background

Star-forming galaxies (SFGs) exhibit characteristic optical emission-line spectra that allow detailed analyses of the physical conditions and chemical abundances of their ionized gas (see Fig. 1). Key diagnostics include the electron temperature and density, which are essential for determining the total oxygen abundance, commonly expressed as 12 + log(O/H) and used as a proxy for gas-phase metallicity or the interstellar medium. Using some essential emission lines, it is possible to determine the chemical abundances of some elements such as oxygen, nitrogen, carbon among others. Oxygen is produced mainly by core-collapse supernovae of massive stars ($M > 8\ M_\odot$), making it a central tracer of galactic chemical enrichment. Nitrogen has a more complex origin, with production channels arising from massive and intermediate-mass stars on different timescales. As such, the nitrogen-to-oxygen (N/O) ratio versus O/H relation is a powerful tool for tracing chemical evolution.

Overall, local SFGs enable more detailed studies than those possible at high redshift. In this context, by examining a sample spanning a broad range of physical conditions, we can investigate how gas properties influence metallicity and key abundance ratios such as the carbon-to-oxygen (%) and N/O ratios

. Recent JWST observations show that high-z galaxies often exhibit N/O ratios that deviate significantly from those of local SFGs, suggesting different nucleosynthetic pathways or conditions, particularly in compact, dense systems. Understanding the role of density in shaping abundances at high redshift therefore requires detailed diagnostics from low-z analogues.

Objective: In this project, we aim to determine the physical conditions and chemical abundances of a sample of local galaxies (z~0) selected to cover the parameter space observed in high-z galaxies. We will use two main literature samples:

1) CLASSY is a treasury survey that builds on the Hubble Space Telescope (HST)/ Cosmic Origins Spectrograph (COS) archive to construct the first high-quality, high-resolution far-ultraviolet (UV) spectral database of 45 nearby Star-Forming Galaxies (PI: Danielle Berg). With these observations, CLASSY will provide the well-controlled local UV+optical sample of galaxies with the sensitivity and resolution to enable synergetics studies of stars and gas within the same galaxies. These galaxies were highly selected to cover similar properties than those galaxies with a higher redshift (z >6).

2) LzLCS, the Lows-redshift Lyman Continuum Survey (LzLCS), comprises a sample of ~ 60 galaxies with spectra from the UV (HST/COS) and optical (SDSS+DESI). This sample was selected to understand the properties and pathways of the Lyman continuum (LyC) photos responsible for the reionization of the Universe. Such study is only possible by analyzing the stellar and gas properties of local star-forming galaxies. With LzLCs, this project will span the range of physical properties not covered by CLASSY.

Proposed work project:

To analyze the ionized gas of 45 CLASSY + ~60 LzLCS galaxies. The student will compute the phase-gas metallicity and the nitrogen abundances for the whole sample in a homogenous way. The student will compare the N/O abundance ratios with different properties of the galaxies (metallicity, ionization parameter, stellar mass, star-formation rate, and selected line ratios). In particular, we will study the N/O versus O/H relation and the variation of this relation with different galaxy properties.

Additional information:

CLASSY: Scientific cases: Berg et al. 2022, ApJ, in press : The COS legacy Spectroscopy SurveY (CLASSY) Treasury atlas.

CLASSY Sample:

For this project, we will focus on the optical spectra from the SDSS sample (38 observations) and the follow-up observations using the VLT/MUSE, LBT/MODS, VLT/KWCI/ESI observations. The total sample comprises 45 star-forming galaxies of the local universe. The SDSS fiber of 3" (diameter) is similar in size to the HST/COS aperture of 2.5" of diameter and so captures similar amounts of UV and optical light.

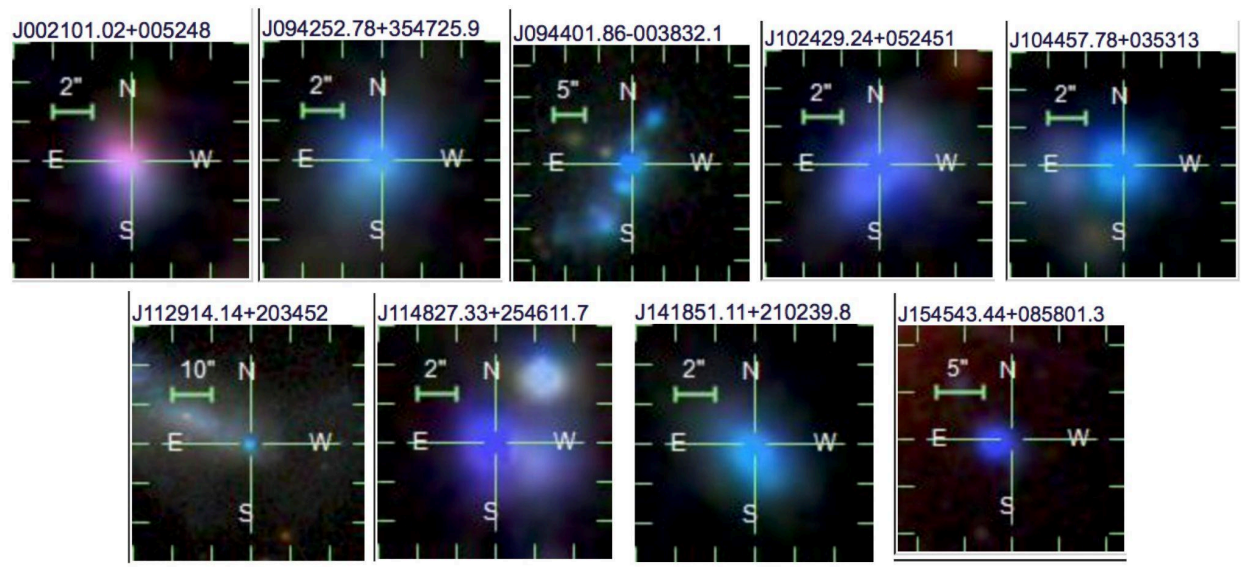

Figure 1. SDSS r-band images of some CLASSY galaxies. These galaxies have a different set of optical observations. For UV images see CLASSY I paper: Berg et al. 2022 (see here).

Preview analysis (CLASSY IV and V paper)

1. The optical spectra are corrected by Galactic extinction.
2. Stellar population subtraction: to correct for underlying stellar population, we have used the STARLIGHT code, which affects mainly recombination lines (e.g., Hβ, Ha and Hγ). We correct for redshift (rest frame wavelength) our sample during this procedure.
3. Fitting emission lines: We fit relevant emission lines of each spectra to get their fluxes and the associated uncertainties. We use the Python task, LMFIT, to fit a single Gaussian to each line. We have fitted the following essential emission lines for this analysis (see Figure 2):

T_e

- [O II] λ3727+3729
- Hδ 4101
- Hγ 4341
- [O III] λ4363 → Temperature-sensitive line (Te (K))
- [Ar IV] 4711, 4741 → Density diagnostic (Ne (cm^-3))
- Hβ 4861
- [O III] λλ4959, 5007
- [N II] λ5755 → Te-sensitive line
- [S III] λ6312 → Te-sensitive line
- [N II] λλ6548, 6584
- Ha 6564
- [S II] λλ6717, 6731 → Ne diagnostic
- [O II] λλ7318+20, 7329+30
- [S III] λ9069

n_e

The measurements of the fluxes will allow us to study the physical conditions (electron density and temperature) and chemical abundances, the oxygen abundance (12+log(O/H)or metallicity), for this sample using the different optical observations. The whole sample has chemical abundance determinations by the Temperature-sensitive method (the direct method).

For Nitrogen, the student will consider:

1. Temperature structure of the nebular
2. The ionization correction factors → we will test at least three most common approaches.

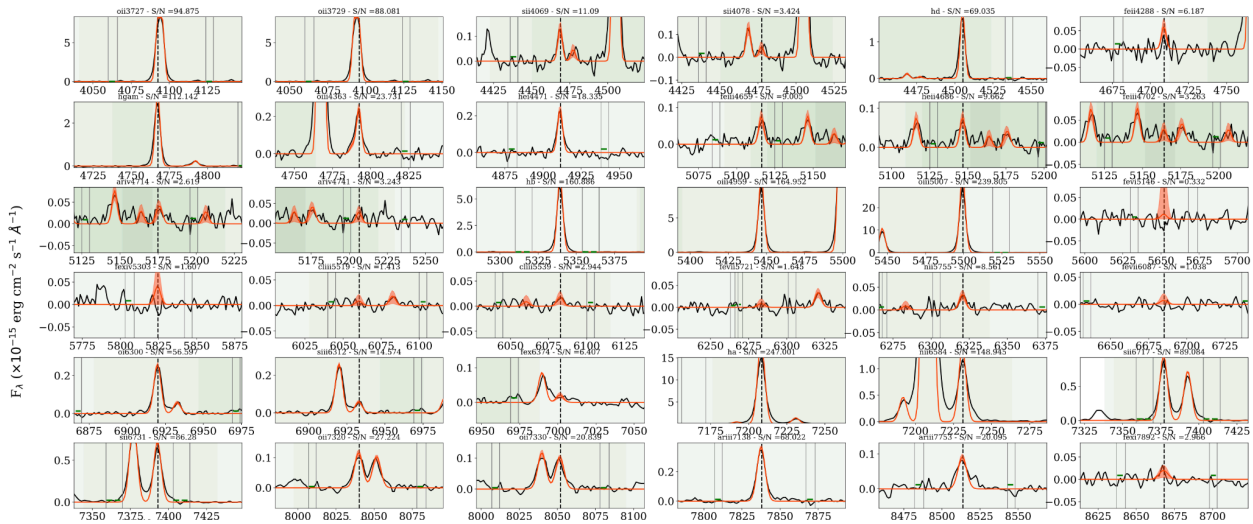

Figure 2. Example of emission spectrum of J0021+0052 of APO/SDSS using an aperture of 3.0". In red Gaussian fitting for different emission lines, mainly, the [O III] λ4363, Hβ, [O III] λλ4959, 5007, [S III] λ6312 [N II], λλ6548, 6584, Ha 6564 and [S II] λλ6717, 6731 emission lines.

Project time:

Week 1: Establish a working relationship. Share information about the background of the project.

- Basic concepts of the the physics of the ionized nebulae
- Sample CLASSY and LzLCS
- Sample description and tools to visualize the spectra

The student and meteor will establish formal meetings in person and via Zoom and communication via mail or TEEMS. We will speak about the motivation of the research project and the goals. The student will start to learn about the physics of ionized objects and emission spectra. We also establish the hour of our meetings. The literature for the first part of the project is here.

Week 2-3: The student will review basic Python tools to display and review the sample of galaxies.

During these weeks, the student will use PyNeb, a tool to analyze the physical properties and chemical composition of the ionized gas. The mentor will explain some specific routines of PyNeb (via Python), and the student will start to do small exercises

including in the manual of PyNeb. The mentor will provide to the student the measurements of the fluxes for the sample to give a first look using Python.

Analysis of the data using Python Jupyter notebooks.

- Reddening correction (To analyze different reddening laws) for LzLCs
- Using PyNeb to analyze the physical conditions (temperature and density) using different diagnostics ([O III] λλ(4959, 5007)/4363 and [S III] λλ(9069, 9534)/6312 for Te and [S II] λλ6717/6731 for density)

Week 4-5:

Student will continue analyzing the data:

- Computation of the uncertainties using Montecarlo for Te and Ne.
- Make preliminary plots to compare the results of electron density and temperature for the different aperture analysis for each galaxy.
- Calculate O+ ionic abundance using either [O II] λ3727+3729 or [O II] λλ7318+20, 7329+30, O++ using the [O III] λλ4959, 5007 emission lines and N+ using [N II] λλ6548, 6584 (PyNeb).
- Ionization correction factor analysis for N/O

Week 6-7:

In these weeks, the student will complete the procedure to obtain the metallicity using the different observations for each galaxy.

- Make preliminary plots between different line ratios
- Make preliminary plots of these results for the set of observations.
- Analysis of the N/O abundance ratio and comparison with galaxy properties (SFR and stellar mass).

Week 8-10:

Final remarks of the N/O abundance ratio. Discussion to select the values of metallicity for these galaxies.